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Bihar Engineering University, Patna

B.Tech 5th Semester Exam 2019

Time: 3 hours

Code: 031506 *Subject: Electromagnetic Field Theory* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Write short answer of the following questions (any seven): **[14]**

(a) Explain the importance of a unit vector.

(b) State divergence theorem and give its mathematical form.

(c) Define propagation constant.

(d) What do you understand by homogeneous and isotropic medium?

(e) Write down Maxwell's equation in free space.

(f) What is dissipation factor of dielectric?

(g) What is displacement current? Give its expression.

(h) What is rotational and irrotational vector field?

(i) Are all the four Maxwell's equations independent? Explain briefly.

(j) Explain briefly the significance of skin depth.

Q.2 Solve all questions : **[14]**

(a) Express the position vector $r = xa_x + ya_y + za_z$ in the spherical coordinate system.

(b) State and prove the Gauss's theorem. Explain why it is called the divergence theorem.

(c) Justify that the net electric field within a conductor is always zero.

Q.3 Solve both questions : **[14]**

(a) Let the spherical surfaces $r = 4$ cm and $r = 9$ cm be separated by two perfect dielectric shells, $\varepsilon_{R1} = 2$ for $4 < r < 6$ cm and $\varepsilon_{R2} = 5$ for $6 < r < 9$ cm. If $E_1 = (2000/r^2)a_r$ V/m, find (a) E_2; (b) the total electrostatic energy stored in each region.

(b) Derive the Laplace equation from Gauss's law in electrostatics.

Q.4 Solve both questions : **[14]**

(a) Derive the Maxwell's curl equation for time varying electric fields.

(b) The magnetic field intensity in a certain conducting medium is $H = xy^2a_x + x^2za_y - y^2z\,a_z$ A/m. Calculate the current density at point $P(2, -1, 3)$. What is dp_v/dt at P?

Q.5 Solve both questions : **[14]**

(a) What is the limitation of Ampere's circuital law? Explain the correction done by Maxwell to Ampere's law by explaining continuity equation.

(b) For a current distribution in free space $A = (2x^2y + yz)a_x + (xy^2 - xz^3)a_y - (6xyz - 2x^2y^2)a_z$ Wb/m. Calculate magnetic flux density.

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Q.6 Solve both questions : **[14]**

(a) A plane electromagnetic wave described by its magnetic field is given by the expression, H= $H_0\sin(kz-\omega t)y$. Determine the corresponding electric field and the time average Poynting vector. If it is incident on a perfect conductor and is totally reflected what would be the pressure exerted on the surface? Determine the surface current generated at the interface.

(b) Derive a wave equation for non-dissipative medium making use of Maxwell equations and field vectors E and H.

Q.7 Solve both questions : **[14]**

(a) A uniform plane wave is incident on the interface of two perfect dielectric media with relative permittivities of ε_1 and ε_2. The electric field E is parallel to the plane of incidence. Show that reflection coefficient $\Gamma = E_r/E_i$ and transmission coefficient $\tau = E_t/E_i$ are given by $\Gamma=[\sqrt{(\varepsilon_2)}\cos\theta_1-\sqrt{(\varepsilon_1)}\cos\theta_2] / [\sqrt{(\varepsilon_2)}\cos\theta_1+\sqrt{(\varepsilon_1)}\cos\theta_2]$; $\tau=[2\sqrt{(\varepsilon_2)}\cos\theta_1] / [\sqrt{(\varepsilon_2)}\cos\theta_1+\sqrt{(\varepsilon_1)}\cos\theta_2]$ where θ_1 and θ_2 are angles of incidence and refraction, respectively.

(b) State Poynting's theorem. What is Poynting vector?

Q.8 Solve both questions : **[14]**

(a) What is the boundary condition? Derive the law of refraction of the electric field at a dielectric-dielectric boundary free of charge conditions.

(b) The transmission line is excited by a voltage source $V_0\cos wt$ at $z=0$. What are the voltage and current distributions if the line is short circuited at $z=l$?

Q.9 Solve both questions : **[14]**

(a) Find the input impedance of the distortion-less transmission line at radio frequencies in both open-circuited and shorted cases.

(b) A 100 ohm line with air dielectric is terminated by a load of $75+j40$ ohm and is excited at 1GHz by a matched generator. Find the position of a single matching stub of 100 ohm impedance on the line and determine the length of the stub.

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